

Mean Time Spent in Transient States

- Consider a finite state Markov Chain with a set of transient states: $T = \{1, 2, \dots, t\}$.

- Gambler's ruin problem:

states: $\{0, 1, \dots, N\}$.

The transient states are $1, 2, \dots, N - 1$.

$$T = \{1, 2, \dots, N - 1\} .$$

- Let the transition probability matrix be $\mathbf{P}$.
- A part of $\mathbf{P}$ formed by probabilities from transient states to transient states:

$$\mathbf{P}_T = \begin{bmatrix} P_{11} & P_{12} & \cdots & P_{1t} \\ P_{21} & P_{22} & \cdots & P_{2t} \\ \vdots & \vdots & \vdots & \vdots \\ P_{t1} & P_{t2} & \cdots & P_{tt} \end{bmatrix}$$

- Examples:

- Gamblers ruin problem:

$$\mathbf{P} = \begin{bmatrix} P_{00} & P_{01} & \cdots & P_{0,N-1} & P_{0,N} \\ P_{10} & P_{11} & \cdots & P_{1,N-1} & P_{1,N} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ P_{N0} & P_{N1} & \cdots & P_{N,N-1} & P_{N,N} \end{bmatrix}$$

$$\mathbf{P}_T = \begin{bmatrix} P_{11} & P_{12} & \cdots & P_{1,N-1} \\ P_{21} & P_{22} & \cdots & P_{2,N-1} \\ \vdots & \vdots & \vdots & \vdots \\ P_{N-1,1} & P_{N-1,2} & \cdots & P_{N-1,N-1} \end{bmatrix}$$

- Suppose a four state MC with states $\{0, 1, 2, 3\}$ has two transient states: 0 and 3. Then:

$$\mathbf{P} = \begin{bmatrix} P_{00} & P_{01} & P_{02} & P_{03} \\ P_{10} & P_{11} & P_{12} & P_{13} \\ P_{20} & P_{21} & P_{22} & P_{23} \\ P_{30} & P_{31} & P_{32} & P_{33} \end{bmatrix} \quad \mathbf{P}_T = \begin{bmatrix} P_{00} & P_{03} \\ P_{30} & P_{33} \end{bmatrix}$$

- For transient states i and j :
 - s_{ij} : expected number of time periods the MC is in state j , given that it starts in state i .
 - Special case s_{ii} : starting from i , the number of time periods in i .
 - Transient states: $f_i < 1$. Recall that f_i is the probability of ever revisit state i starting from state i .
 - Define f_{ij} : the probability that the MC ever visits state j given that it starts in i .
Special case: $f_{ii} = f_i$.

Compute s_{ij} by conditioning:

If $i = j$:

$$s_{ij} = 1 + E\{\# \text{ visits to } j \text{ after the initial state}\} \quad (1)$$

If $i \neq j$:

$$s_{ij} = E\{\# \text{ visits to } j \text{ after the initial state}\} \quad (2)$$

Note

$$\begin{aligned} & E\{\# \text{ visits to } j \text{ after the initial state}\} \\ &= \sum_k P\{\text{the first transition is to state } k\} \times \\ & \quad E\{\# \text{ visits to } j \mid \text{starting from } k\} \\ &= \sum_k P_{ik} s_{kj} \end{aligned}$$

If k is recurrent, then $P_{kj}^n = 0$ for all n . Hence $s_{kj} = 0$. To see $P_{kj}^n = 0$ for all n , suppose $P_{kj}^n > 0$ for some n . Then $P_{jk}^m = 0$ for all m , otherwise k and j communicate and this conflicts the fact j is transient. Since $P_{jk}^m = 0$ for all m , with a positive probability state k will transit to state j and never come back to state k , which conflicts the fact k is recurrent (k should be revisited infinitely many times). Hence $P_{kj}^n > 0$ for some n is not true.

The above equation can be written as:

$$E\{\# \text{ visits to } j \text{ after the initial state}\} = \sum_{k=1}^t P_{ik} s_{kj}$$

To combine Eq. (1) and (2) into a unified form:

$$\delta_{i,j} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Hence:

$$s_{ij} = \delta_{ij} + \sum_{k=1}^t P_{ik} s_{kj} \quad (3)$$

for all $i, j \in \{1, 2, \dots, t\}$.

Use matrix:

$$S = \begin{bmatrix} s_{11} & s_{12} & \cdots & s_{1t} \\ s_{21} & s_{22} & \cdots & s_{2t} \\ \vdots & \vdots & \vdots & \vdots \\ s_{t1} & s_{t2} & \cdots & s_{tt} \end{bmatrix}$$

Eq. (3) is equivalent to

$$\begin{aligned} S &= I + \mathbf{P}_T S \\ S &= (I - \mathbf{P}_T)^{-1} \end{aligned}$$

To obtain f_{ij} , the probability of ever transit to j starting in i , compute expectation by conditioning:

$$\begin{aligned}
 s_{ij} &= P\{\text{ever transit to } j \mid \text{start in } i\} \times \\
 &\quad E[\text{time spent in } j \mid \text{start in } i \text{ and transit to } j] + \\
 &\quad P\{\text{never transit to } j \mid \text{start in } i\} \times \\
 &\quad E[\text{time spent in } j \mid \text{start in } i, \text{ never transit to } j] \\
 &= f_{ij}(\delta_{ij} + s_{jj}) + (1 - f_{ij})\delta_{ij} \\
 &= \delta_{ij} + f_{ij}s_{jj}
 \end{aligned}$$

Hence

$$f_{ij} = \frac{s_{ij} - \delta_{ij}}{s_{jj}} \quad (4)$$

Special case:

$$f_{ii} = \frac{s_{ii} - 1}{s_{ii}} .$$

Example: Gambler's ruin problem. Suppose $p = 0.4$, $N = 7$. Start with 3. (a) The expected amount of time the gambler has 5 units, or 2 units. (b) The probability that the gambler ever has a fortune of 1.

Solution:

States: $\{0, 1, 2, 3, 4, 5, 6, 7\}$.

Transient states: $\{1, 2, 3, 4, 5, 6\}$.

$$\mathbf{P}_T = \begin{bmatrix} 0 & 0.4 & 0 & 0 & 0 & 0 \\ 0.6 & 0 & 0.4 & 0 & 0 & 0 \\ 0 & 0.6 & 0 & 0.4 & 0 & 0 \\ 0 & 0 & 0.6 & 0 & 0.4 & 0 \\ 0 & 0 & 0 & 0.6 & 0 & 0.4 \\ 0 & 0 & 0 & 0 & 0.6 & 0 \end{bmatrix}$$

Solve $S = (I - \mathbf{P}_T)^{-1}$, we get

$$s_{3,5} = 0.9228 \quad s_{3,2} = 2.3677$$

Hence the expected number of times the gambler has 5 units is 0.9228, 2 units is 2.3677.

$f_{3,1} = \frac{s_{3,1} - \delta_{3,1}}{s_{1,1}} = \frac{s_{3,1}}{s_{1,1}} = \frac{1.4206}{1.6149} = 0.8797$. The probability that the gambler ever has a fortune of 1 is 0.8797.